Fairness Certificates via a Lean-Backed Probabilistic Typed Deduction

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Abstract

We present *TPTND-Lean*, a Lean 4 implementation of a core fragment of Trustworthy Probabilistic Typed Natural Deduction (TPTND). TPTND-Lean type-checks derivation trees whose leaves are empirical observations and whose internal nodes are introduction or elimination rules for probabilistic connectives, producing machine-checked *fairness certificates*. Counts, frequencies, and stored interval endpoints are exact rational; the standard-error term used by the score-test acceptance bands is computed by a bounded rational Newton iteration. We extend the original one-sample trust rules with two-sample variants IT2/IUT2 that compare two observed populations directly, and we describe the four-stage workflow that takes group-level summary statistics through claim selection and derivation construction to a checked certificate. Two case studies on public data illustrate the approach: (i) the ProPublica COMPAS recidivism data (6 172 defendants, reproducing the headline racial-bias finding), and (ii) HMDA mortgage-lending data for Delaware (54 000 originated-or-denied Black/White applications across 2022–2023, demonstrating multi-year temporal composition via UPDATE and intersectional comparisons). Certificates are *conditional*: they verify the derivation relative to the supplied counts, the chosen statistical rule, and the checker, not the upstream data pipeline. The full source, data pipelines, and all derivations are publicly available at https://github.com/dasaro/tptnd-lean.

Keywords

algorithmic fairness, typed natural deduction, proof checking, Lean 4, trust certificates

1. Introduction

Automated decision-making systems increasingly affect high-stakes outcomes (criminal sentencing, credit scoring, mortgage lending), and the question of whether their outputs are *fair* across demographic groups has become a central concern in AI ethics [8]. Statistical tests for group-fairness properties (equalized odds, calibration, disparate impact) are routinely applied in auditing practice, yet the resulting claims are typically embedded in narrative reports or Jupyter notebooks, with no formal guarantee that the stated assumptions, data, and statistical procedures are consistent.

Trustworthy Probabilistic Typed Natural Deduction (TPTND) [1] offers a proof-theoretic alternative. TPTND is a natural-deduction calculus whose types are probability distributions over finite output alphabets and whose terms carry sample-size and provenance annotations. Introduction rules for *trust* and *untrust* internalize a confidence-interval membership test: if the model probability p lies inside the interval $[\ell, h]$ computed from observed data, the calculus derives **Trust**; otherwise **UTrust**. Elimination rules extract the interval into the typing context, enabling further composition.

In this paper we describe **TPTND-Lean**, a Lean 4 implementation of a core TPTND fragment. Our contributions are:

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1. A **machine-checkable implementation** of the calculus (Section 2.5), where every rule is a pure function |Derivation → CheckM Unit| dispatched by rule name, and all arithmetic is exact rational.
2. **Two-sample trust rules** IT2 and IUT2 (Table 1), which compare two observed populations without designating either as a fixed benchmark.
3. A **four-stage auditing workflow** that takes group-level summary statistics through claim selection and derivation construction to a machine-checked certificate (Section 3.4).
4. **Two case studies** on public data (Section 4): COMPAS recidivism scoring and HMDA mortgage lending, producing derivation trees of depth up to 3 that include intersectional and year-over-year comparisons.

2. The TPTND Calculus

Background. We treat each atomic output a as an event in some implicit finite sample space, with $P(a)$ its marginal probability. The output algebra mirrors standard event operations: $\overline{\alpha}$ is the complement, $\alpha + \beta$ is disjoint union (the calculus enforces $\alpha \perp_{\text{syn}} \beta$ syntactically, so $P(\alpha + \beta) = P(\alpha) + P(\beta)$), $\alpha \times \beta$ is the joint event under an explicit independence witness ($P(\alpha \times \beta) = P(\alpha)P(\beta)$), and $\alpha \Rightarrow_p \beta$ encodes the conditional $P(\beta \mid \alpha) = p$. Two modalities of judgement accompany distributions: *expected mode* $t_n : \alpha_{\tilde{p}}$ records a model-side prediction that the fraction of n samples falling in α is $\tilde{p}$; *frequency mode* $t_n : \alpha_f$ records the empirical fraction f. The trust rules reconcile the two via score-test acceptance bands $\mathcal{P}(n, f, p)$.

We summarize the syntax, judgement forms, and key rules. The full calculus (31 rules across 7 families) appears in [1]; here we present only the rules central to the case studies: Obs, Update, and the trust family (Table 1). Our main extension is two-sample trust (IT2/IUT2).

2.1. Syntax

Outputs are generated by $\alpha, \beta ::= a \mid \overline{\alpha} \mid \alpha + \beta \mid \alpha \times \beta \mid \alpha \Rightarrow_p \beta$, where a ranges over a countable set of atomic names and $p \in [0, 1]$ is the conditional probability annotation on arrows.

Probability constraints are $c ::= p \mid [\ell, h] \mid \neg[\ell, h]$, with $p, \ell, h \in [0, 1]$ and $\ell \leq h$. The constraint $\neg[\ell, h]$ records that the relevant probability lies *outside* the interval; it is introduced by EUT.

Terms are $t ::= a \mid \langle t, u\rangle \mid \mathsf{fst}(t) \mid \mathsf{snd}(t) \mid [x]t \mid (t \cdot u)$.

A *context* Γ is a finite list of assumptions $x : \alpha_c$ equipped with a nonempty support annotation. Term sequents carry an explicit provenance index σ. For product and structural rules we write $\Gamma \#_w \Delta$ when the derivation carries an explicit independence witness w.

2.2. Judgement forms

The calculus uses ten sequent forms [1]:

$\vdash \alpha :: \mathsf{output}$	output declaration
$\vdash \Gamma$	distribution declaration
$\Gamma \vdash x : \alpha_c$	identity (exact model import)
$\Gamma \vdash_\sigma t : \alpha$	experiment-mode atomic claim
$\Gamma \vdash_\sigma t_n : \alpha_{\tilde{p}}$	expected-mode claim
$\Gamma \vdash_\sigma t_n : \alpha_f$	frequency-mode claim
$\Gamma \vdash \mathbf{Trust}_{\mathcal{P}}(t_n : \alpha_f; p, [\ell, h])$	trust certificate
$\Gamma \vdash \mathbf{UTrust}_{\mathcal{P}}(t_n : \alpha_f; p, [\ell, h])$	untrust certificate
$\Gamma \vdash \mathbf{Excess}_{\mathcal{Q}}(t_n : \alpha_f \succ u_m : \alpha_g; d, [\ell, h])$	excess certificate
$\Gamma \vdash \mathbf{NoExcess}_{\mathcal{Q}}(t_n : \alpha_f \succ u_m : \alpha_g; d, [\ell, h])$	no-excess certificate

The one-sample acceptance band for the proportion z-test of a fixed null p is $\mathcal{P}(n, f, p) = f \pm z_{.975}\sqrt{p(1-p)/n}$, clamped to $[0, 1]$, where $z_{.975} = 1.96$; the trust rules use $p \in \mathcal{P}(n, f, p)$ as the test condition. The two-sample band, used by IT2/IUT2 to test $f{=}g$, is $\mathcal{Q}(n,m,f,g) = (f-g) \pm z_{.975}\sqrt{\hat{p}(1-\hat{p})(1/n+1/m)}$, with $\hat{p} = (nf+mg)/(n+m)$ the pooled rate under $H_0 : p_1{=}p_2$. We continue to write $\mathcal{P}/\mathcal{Q}$ for these bands but they are test-acceptance regions, not classical confidence intervals.

2.3. Global admissibility

Before any rule fires, the conclusion context must satisfy $\mathsf{wf}(\Gamma)$: every constraint is well-formed ($\ell \leq h$), every support is nonempty, and for each variable the sum of exact masses and interval lower bounds does not exceed 1.

2.4. Running example

Consider a simplified lending audit. A context Γ declares an output type Denied and a benchmark denial rate of 20%: $x : \mathsf{Denied}_{0.2}$. An auditor observes $n = 100$ applicants from group A, of whom 35 are denied, recording this via Obs:

$$\Gamma \vdash_{\{src_A\}} t_{100} : \mathsf{Denied}_{0.35}$$

To test whether this observed rate is consistent with the 20% benchmark, we apply IUT. The one-sample CI is $\mathcal{P}(100, 0.35, 0.2) = 0.35 \pm 1.96\sqrt{0.2 \cdot 0.8/100} = [0.272, 0.428]$. Since $0.2 \notin [0.272, 0.428]$, the calculus derives:

$$\Gamma \vdash \mathbf{UTrust}_{\mathcal{P}}(t_{100} : \mathsf{Denied}_{0.35};\ 0.2,\ [0.272, 0.428])$$

This certificate is a self-contained artifact: anyone with the checker can verify that the derivation is valid without re-running the statistical test or having access to the raw data. The next section presents the rules formally.

2.5. Key rules

Observation (Obs). The fundamental data-entry rule: given a context entry supporting term t, record n observations with empirical frequency f:

$$\frac{\sigma \neq \varnothing \quad \exists!\, x{:}\alpha_c \in \Gamma \text{ supporting } t \quad n > 0 \quad nf \in \mathbb{N}}{\Gamma \vdash_\sigma t_n : \alpha_f}\ \textsc{Obs}$$

The provenance σ (a set of data-source identifiers) tracks which observations contribute to a claim.

Update (Update). Pools two frequency claims with disjoint provenance into a combined weighted frequency:

$$\frac{\Gamma \vdash_{\sigma_1} t_n : \alpha_f \quad \Gamma \vdash_{\sigma_2} t_m : \alpha_g \quad \sigma_1 \cap \sigma_2 = \varnothing}{\Gamma \vdash_{\sigma_1 \uplus \sigma_2} t_{n+m} : \alpha_{\frac{nf+mg}{n+m}}}\ \textsc{Update}$$

Identity (Identity). Imports a model probability from the context: $\dfrac{x : \alpha_p \in \Gamma}{\Gamma \vdash x : \alpha_p}$.

The remaining rules (output formation, products, arrows, sum types, structural weakening, and contraction) appear in [1] and are implemented in TPTND-Lean but not exercised in the case studies below. The Bayesian-update rules I-P and E-P from [1] are also implemented and are exercised once in the COMPAS case study (Section 4.1, Table 4).

$$\frac{\Gamma \vdash x : \alpha_p \quad \Delta \vdash_\sigma u_n : \alpha_f \qquad \mathcal{P}(n,f,p) = [\ell,h] \quad p \in [\ell,h]}{\Gamma, \Delta \vdash \textsf{\textbf{Trust}}_{\mathcal{P}}(u_n : \alpha_f; p, [\ell,h])}\ \mathrm{IT}$$

$$\frac{\Gamma \vdash x : \alpha_p \quad \Delta \vdash_\sigma u_n : \alpha_f \qquad \mathcal{P}(n,f,p) = [\ell,h] \quad p \notin [\ell,h]}{\Gamma, \Delta \vdash \textsf{\textbf{UTrust}}_{\mathcal{P}}(u_n : \alpha_f; p, [\ell,h])}\ \mathrm{IUT}$$

$$\frac{\Gamma \vdash_\sigma t_n : \alpha_f \quad \Delta \vdash_\tau u_m : \alpha_g \qquad \sigma \cap \tau = \varnothing \qquad \mathcal{Q}(n,m,f,g) = [\ell,h] \quad 0 \in [\ell,h]}{\Gamma, \Delta \vdash \textsf{\textbf{Trust}}_{\mathcal{Q}}(t_n : \alpha_f; g, [\ell,h])}\ \mathrm{IT2}$$

$$\frac{\Gamma \vdash_\sigma t_n : \alpha_f \quad \Delta \vdash_\tau u_m : \alpha_g \qquad \sigma \cap \tau = \varnothing \qquad \mathcal{Q}(n,m,f,g) = [\ell,h] \quad 0 \notin [\ell,h]}{\Gamma, \Delta \vdash \textsf{\textbf{UTrust}}_{\mathcal{Q}}(t_n : \alpha_f; g, [\ell,h])}\ \mathrm{IUT2}$$

Table 1
Trust-introduction rules. IT/IUT are one-sample (model p is a known constant). $\mathrm{IT2}/\mathrm{IUT2}$ (new in this work) are two-sample: both f and g are estimated, and the test is whether 0 lies inside the CI for $f-g$.

$$\frac{\Gamma \vdash \textsf{\textbf{Trust}}_{\mathcal{P}}(u_n : \alpha_f; p, [\ell,h])}{\Gamma, x_u : \alpha_{[\ell,h]} \vdash_\sigma u_n : \alpha_f}\ \mathrm{ET} \qquad \frac{\Gamma \vdash \textsf{\textbf{UTrust}}_{\mathcal{P}}(u_n : \alpha_f; p, [\ell,h])}{\Gamma, x_u : \alpha_{\neg[\ell,h]} \vdash_\sigma u_n : \alpha_f}\ \mathrm{EUT}$$

$$\frac{\Gamma, x_u : \alpha_c \vdash \textsf{\textbf{Trust}}_{\mathcal{P}}(u_n : \alpha_f; p, [\ell,h]) \qquad p \in c}{\Gamma, x_u : \alpha_p \vdash_\sigma u_n : \alpha_{\tilde{p}}}\ \mathrm{ET}^{\mathrm{EX}}$$

$$\frac{\Gamma \vdash_\sigma t_n : \alpha_f \quad \Delta \vdash_\tau u_m : \alpha_g \qquad \sigma \cap \tau = \varnothing \quad \mathcal{Q}(n,m,f,g) = [\ell,h] \quad 0 \notin [\ell,h]}{\Gamma, \Delta \vdash \textsf{\textbf{Excess}}_{\mathcal{Q}}(t_n : \alpha_f \succ u_m : \alpha_g;\ f-g,\ [\ell,h])}\ \mathrm{IE}\textsc{x}$$

$$\frac{\Gamma \vdash_\sigma t_n : \alpha_f \quad \Delta \vdash_\tau u_m : \alpha_g \qquad \sigma \cap \tau = \varnothing \quad \mathcal{Q}(n,m,f,g) = [\ell,h] \quad 0 \in [\ell,h]}{\Gamma, \Delta \vdash \textsf{\textbf{NoExcess}}_{\mathcal{Q}}(t_n : \alpha_f \sim u_m : \alpha_g;\ f-g,\ [\ell,h])}\ \mathrm{INE}\textsc{x}$$

Table 2
Trust elimination, comparison, and comparison-elimination rules. $\mathrm{EE}\textsc{x}$ and $\mathrm{ENE}\textsc{x}$ (not shown) re-enter the frequency layer under an excess or no-excess certificate, shifting the left observation by a model-derived interval.

3. TPTND-Lean: Implementation in Lean 4

3.1. Kernel

TPTND-Lean is approximately 1 700 lines of Lean 4: a read-only kernel (probability arithmetic, well-formedness, and one checker per rule) surfaced through a single entry point, `checkDerivation` : *Derivation* → `Except String Unit`, which traverses a derivation tree and either returns successfully, in which case the tree *is* the certificate, or yields an error message naming the failing premise. Three design choices reduce the trusted surface of the checker. A probability is wrapped in a `Prob` structure carrying proof witnesses of $0 \leq q \leq 1$ at construction time, so an ill-formed probability value is unrepresentable. The checking monad forces every rule to return either a checked result or an explicit error, ruling out silent failures. Rule names are a closed inductive datatype, so the dispatcher is exhaustively type-checked and a misspelled rule is a compile-time error. We do not yet prove a meta-theorem connecting checker success to an independently defined semantic validity relation: the current trusted base consists of the Lean compiler, the `partial`-declared dispatcher, the rational square-root approximation (Section 3.2), and the external pipeline that produces the leaf counts.

3.2. Rational arithmetic and provenance

All arithmetic is exact rational ($\mathbb{Q}$); the $[0,1]$ bounds are enforced as type-level proof obligations on every probability value.

Square roots for the CI standard-error term are approximated by a rational Newton iteration (15 steps), followed by rounding to six decimal places to keep denominators bounded. The z-value for the 95% two-sided interval is the rational $49/25 = 1.96$. The square-root rounding is the one place where TPTND-Lean is not bit-exact: the rounded value can fall on either side of the true $\sqrt{}$ by up to $5 \cdot 10^{-7}$, so each CI endpoint is accurate to within $\approx z \cdot 5 \cdot 10^{-7} \approx 10^{-6}$. A derivation in which the model probability p falls within this tolerance of an interval boundary is therefore not safely decidable by TPTND-Lean; we treat it as a known soundness boundary rather than a guaranteed verdict, and none of the case-study derivations sit in this regime.

Provenance and probability conservation. The provenance index σ has two regimes: I+, E$+_{L,R}$, I$\times$, E$\times_{L,R}$, E$\to$ require all three claims to share σ (disjoint events on one batch); Update, IT2/IUT2, IEx/INEx require the two premise provenances to be *disjoint* with the conclusion's σ their union (independent batches). Without this discipline the textbook attack of [1],

$$\frac{\Gamma \vdash_{\sigma_A} t_4 : \alpha_{3/4} \quad \Gamma \vdash_{\sigma_B} t_4 : \beta_{3/4}}{\Gamma \vdash_{?} t_4 : (\alpha + \beta)_{3/2}} \; \mathrm{I+}$$

would derive a probability of 3/2. In TPTND-Lean the rule layer rejects the $\sigma_A \neq \sigma_B$ premises before any arithmetic runs, and as a backstop the value-level primitives `probAdd`, `probSub`, `probDiv` and `weightedFreq` return `Option Prob` and fail whenever a result would fall outside $[0,1]$. A same-σ attack with $p + q > 1$ is blocked by the value layer; a mismatched-σ attack with $p + q \leq 1$ would slip past the value layer and is caught only by the rule layer.

3.3. Two-sample trust

The original calculus [1] provides one-sample rules IT/IUT, which compare an observed frequency against a known model probability. In practice, however, the "model" is often itself estimated from data (e.g. the White denial rate is not a physical constant but a sample statistic). We add IT2 and IUT2 (Table 1), which take two frequency observations as premises and use the pooled score-test interval $\mathcal{Q}$. The test is: if $0 \in \mathcal{Q}(n, m, f, g)$ then **Trust** (no significant difference); otherwise **UTrust**. The two-sample rules require disjoint provenances $\sigma \cap \tau = \varnothing$, preventing a group from being compared with (a subset of) itself.

3.4. User workflow: producing a fairness certificate

An auditor produces a TPTND-Lean certificate in four stages.

1. Group-level summary statistics. The auditor extracts sample sizes and event counts for the demographic strata of interest. No TPTND-Lean-specific tooling is required at this stage: SQL, pandas, or R suffice. For COMPAS (Section 4.1) the inputs are sample sizes and high-risk counts for each (race, sex, recidivism-status) cell; for HMDA (Section 4.2) they are denial counts indexed by year, race, and sex. This is the only point at which the auditor needs raw data.

2. Claim selection. The fairness question fixes which TPTND-Lean rule closes the derivation. Common patterns:

- *observed rate is consistent with a known benchmark* p $\to$ IT / IUT;
- *two observed populations are statistically indistinguishable* $\to$ IT2 / IUT2;
- *rate increased significantly between two periods* $\to$ IUT2 on year-stratified Obs leaves;
- *the effect persists after pooling sub-cohorts* $\to$ Update chained with one of the above.

The English statement of the audit determines the rule signature unambiguously, with IT/IUT/IT2/IUT2 playing the role that classical hypothesis tests play in a statistical-software workflow.

3. Derivation construction. The derivation tree mirrors the structure of the auditor's English description. Each leaf is an Obs node carrying one (n, f, σ) triple from stage 1; internal nodes apply the rules from stage 2. Rule arity, output-type matching, and provenance disjointness are fixed by the rule, leaving only the data values as free parameters.

4. Verification and certification. A single call to `checkDerivation` traverses the tree, applies the per-rule checker at each node, and returns either `Ok` (the derivation *is* the certificate) or an error message that identifies the failing premise. The certificate is a self-contained inductive value: its validity is independent of the data and tools used to produce it, and any party reconstructing the derivation has the certificate. Every case-study derivation re-checks in well under a second.

The kernel covers stages 2–4; stage 1 remains the auditor's responsibility. A tabular-data front-end and a certificate serialization format are natural next steps.

3.5. Fairness measures expressible in TPTND

TPTND directly encodes any group-fairness criterion that reduces to a difference-of-proportions test on group-conditional event probabilities. In the standard taxonomy [8]:

- **Statistical parity** ($P(\hat{Y}{=}1 \mid A{=}a)$ equal across groups): encoded as IT2/IUT2 on raw acceptance rates; Tree A in Section 4.2 is a one-sample variant against a fixed reference.
- **Equalized odds** [9] ($P(\hat{Y}{=}1 \mid A{=}a, Y{=}y)$ equal across a for each y): encoded as IT2/IUT2 on the Y-conditioned subsets. The ProPublica headline derivation (Section 4.1) is the false-positive case ($y{=}0$); the *equal-opportunity* relaxation [9] uses only $y{=}1$.
- **Predictive parity / calibration** [10] ($P(Y{=}1 \mid \hat{Y}{=}1, A{=}a)$ equal across a): encoded as IT2 on the $\hat{Y}$-conditioned subsets. The certificates produced by TPTND-Lean should not be read as calibration claims: under unequal base rates and a non-perfect predictor, calibration / predictive parity and equalized odds cannot both hold [10], so a COMPAS-style audit must make explicit which fairness property is being certified.

Two important families fall outside this fragment. *Disparate-impact ratio* (an 80% rule on $P(\hat{Y}{=}1 \mid A{=}0)/P(\hat{Y}{=}1 \mid A{=}1)$) is multiplicative and would require an Excess-style family with division. *Individual* and *counterfactual* fairness require causal structure, addressed by the causal-label extensions of TPTND [3, 4].

4. Case Studies

What the certificates verify. The certificates produced below are *conditional*. The checker verifies that the supplied counts and the chosen rule form a valid TPTND derivation; it does not certify that the upstream data extraction (filtering, demographic encoding, outcome construction, hypothesis selection) was done correctly. Those steps remain part of the external audit trail. We also inherit the standard caveats of the underlying statistical tests (binomial sampling, independence, large-sample approximation, no multiple-testing correction).

4.1. COMPAS: recidivism risk scoring

The first case study uses the ProPublica COMPAS dataset [6] (`compas-scores-two-years.csv`, filtered to 6 172 defendants by the published criteria). We construct 12 Lean-validated derivation trees, including:

ProPublica's headline (UTrust). ProPublica's article reports false-positive rates of 44.9% for Black and 23.5% for White non-recidivists [6]. On the published filter (`days_b_screening_arrest` $\in$

$[-30, 30]$, `is_recid` $\neq -1$, `c_charge_degree` $\neq$ `O`, `score_text` $\neq$ `N/A`, yielding 6 172 cases) the corresponding observed rates are $641/1514 \approx 42.3\%$ for Black and $282/1281 = 94/427 \approx 22.0\%$ for White, preserving the $\sim 1.9\times$ disparity. The White FPR is itself a sample statistic, so the statistically more appropriate comparison uses IUT2; we re-derive the two-sample variant in the HMDA case study below (Section 4.2, Trees B and C) and present the IUT version here only for continuity with ProPublica's fixed-benchmark framing. Let Γ_{BN} carry the non-exact assumption $\mathsf{HighRisk}_{[0,1]}$ for the Black cohort and $\Gamma^{\mathrm{mdl}}_{WN} \vdash x_{WN} : \mathsf{HighRisk}_{94/427}$ import the White benchmark; write $\Theta_{FP} = \Gamma^{\mathrm{mdl}}_{WN}, \Gamma_{BN}$ and $\sigma_B = \sigma_m \uplus \sigma_f$. The TPTND-Lean-verified derivation is:

$$\dfrac{\Gamma_{BN} \vdash_{\sigma_m} u_{1168} : \mathsf{HighRisk}_{255/584} \qquad \Gamma_{BN} \vdash_{\sigma_f} u_{346} : \mathsf{HighRisk}_{131/346}}{\Gamma_{BN} \vdash_{\sigma_B} u_{1514} : \mathsf{HighRisk}_{641/1514}}\ \textsc{Update}$$

$$\dfrac{\Gamma_{BN} \vdash_{\sigma_B} u_{1514} : \mathsf{HighRisk}_{641/1514} \qquad \begin{array}{c}\Gamma^{\mathrm{mdl}}_{WN} \vdash x_{WN} : \mathsf{HighRisk}_{94/427} \\ 94/427 \notin \mathcal{P}(1514,\ 641/1514,\ 94/427) = [0.4025, 0.4443]\end{array}}{\Theta_{FP} \vdash \mathbf{UTrust}_{\mathcal{P}}(u_{1514} : \mathsf{HighRisk}_{641/1514};\ 94/427,\ [0.4025, 0.4443])}\ \mathrm{IUT}$$

$$\dfrac{\Theta_{FP} \vdash \mathbf{UTrust}_{\mathcal{P}}(u_{1514} : \mathsf{HighRisk}_{641/1514};\ 94/427,\ [0.4025, 0.4443])}{\Theta_{FP}, x_{BN} : \mathsf{HighRisk}_{\neg[0.4025, 0.4443]} \vdash_{\sigma_B} u_{1514} : \mathsf{HighRisk}_{641/1514}}\ \mathrm{EUT}$$

The resulting term claim, with $\mathsf{HighRisk}_{\neg[0.4025,0.4443]}$ recorded in its typing context, is what a regulator would receive and re-check.

Bayesian belief update (depth 2). A pilot auditor entertaining three hypotheses about the FPR (a_1=1/6, a_2=1/3, a_3=1/2) with prior weights b_i=a_i (summing to 1) observes s=2 flagged among n=5 pilot defendants. I-P introduces the prior family from three exact identity premises; E-P produces the posterior under hypothesis H_j:

$$P(H_j \mid d) = \frac{a_j^s (1-a_j)^{n-s}\, b_j}{\sum_i a_i^s (1-a_i)^{n-s}\, b_i}.$$

For H_3 this gives $729/1366 \approx 53.4\%$ (up from 50%) while H_1 drops to $125/1366 \approx 9.2\%$. The computation is exact rational and the conclusion is a verified posterior usable as a model premise in a downstream IT.

A genuine non-finding (Trust). Among Black recidivists, the female lower-risk rate is 62/203 and the male benchmark is $411/1458 = 137/486$. Since $\mathcal{P}(203, 62/203, 137/486) = [0.2435, 0.3673]$ and $137/486 \approx 0.2819 \in [0.2435, 0.3673]$, the calculus derives **Trust** and $\mathrm{ET}^{\mathrm{EX}}$ re-enters the expected layer with the trusted exact value.

4.2. HMDA: mortgage lending fairness

The second case study uses Home Mortgage Disclosure Act (HMDA) data for Delaware, years 2022 and 2023, downloaded from the CFPB Data Browser.[1] HMDA is a federally mandated dataset under Regulation C (12 CFR 1003), making the "transferable certificate" framing directly grounded in actual regulatory practice.

After restricting to originated (`action_taken` = 1) and denied (`action_taken` = 3) applications, we obtain the summary in Table 3.

We build six derivation trees, organised in four cases that each highlight a different aspect of the certificate framework:

Tree A: reference-rate trust (depth 2). An Identity leaf pins a 20% reference denial rate; an Obs leaf records the White DE 2022 frequency (4914/24614). IT derives **Trust**: the observed rate is consistent with the reference. Every assumption (reference value, data source, sample size, statistical test) is explicit in the certificate and machine-checkable.

[1] https://ffiec.cfpb.gov/data-browser/?category=states&items=DE

Year	Group	Denied	Total	Rate
2022	White	4 914	24 614	19.96%
	Black	2 409	7 142	33.73%
2023	White	3 908	17 355	22.52%
	Black	2 035	5 394	37.73%
2023	Black Male	741	1 825	40.60%
	White Male	1 466	5 450	26.90%
	Black Female	909	2 341	38.83%
	White Female	1 234	4 634	26.63%

Table 3
HMDA Delaware denial rates, 2022–2023.

Tree B: temporal composition (depth 3). Two Obs leaves per racial group (one for 2022, one for 2023) are combined via Update into pooled frequencies. The derivation structure is:

$$\dfrac{\dfrac{\Gamma \vdash_{\sigma_{22}} t : \mathsf{Denied}_{2409/7142} \quad \Gamma \vdash_{\sigma_{23}} t : \mathsf{Denied}_{2035/5394}}{\Gamma \vdash_{\sigma_B} t_{12536} : \mathsf{Denied}_{4444/12536}}\,\textsc{Upd.} \qquad \dfrac{\Gamma \vdash_{\tau_{22}} u : \mathsf{Denied}_{4914/24614} \quad \Gamma \vdash_{\tau_{23}} u : \mathsf{Denied}_{3908/17355}}{\Gamma \vdash_{\tau_W} u_{41969} : \mathsf{Denied}_{8822/41969}}\,\textsc{Upd.}}{\Gamma \vdash \mathbf{UTrust}_{\mathcal{Q}}(\ldots; [0.1357, 0.1528])}\,\text{IUT2}$$

Since $0 \notin [0.1357, 0.1528]$, it yields **UTrust**. The $\mathbf{UTrust}_{\mathcal{Q}}$ certificate is itself transferable: a regulator with the kernel can re-check the entire depth-3 chain (Obs→Update→IUT2) without access to the raw data.

Trees C.1/C.2: intersectional certificates (depth 2). Two auditors independently certify Black-vs-White denial gaps for males (741/1825 vs. 1466/5450, $\mathcal{Q} = [0.1126, 0.1614]$) and females (909/2341 vs. 1234/4634, $\mathcal{Q} = [0.0990, 0.1449]$); both derive **UTrust**. A regulator verifies both certificates with the same kernel without needing access to either auditor's raw data.

Trees D.1/D.2: year-over-year monitoring (depth 2). Black denial rose 2022→2023 from 33.7% to 37.7% ($\mathcal{Q} = [0.0230, 0.0568]$, **UTrust**); White from 20.0% to 22.5% ($\mathcal{Q} = [0.0176, 0.0334]$, **UTrust**); both increases are statistically significant. Persistent national-level racial disparities in mortgage lending are documented by the Urban Institute [7]. Dated certificates accumulate in a regulatory audit trail.

4.3. Summary

Table 4 summarizes all Lean-validated derivation trees across both case studies.

5. Related Work

TPTND was introduced by D'Asaro, Genco, and Primiero [1]; a companion λ-calculus formulation appears in [2], and extensions with causal labels by Ceragioli and Primiero [3, 4]. The Brio tool [5] evaluates bias via distance metrics but does not produce derivation trees. The certifying-algorithms pattern of separating producer from verifier is classical [11, 12]. VeriFair [13] and FairSquare [14] verify fairness properties *a priori* (of the model); TPTND-Lean instead produces *a posteriori* certificates from observed data, checkable without model access. Lean 4 [15] has been used for large-scale mathematical formalization but, to our knowledge, not previously for algorithmic-fairness auditing. Statistical toolkits such as Fairlearn [16], AIF360 [17], and What-If [18] compute metrics but do not produce formally verified certificates: TPTND-Lean's architectural difference is the separation of the (potentially biased) *producer* from the (small, auditable) *checker*.

Dataset	Derivation	Rules	Depth	Result
COMPAS	Caucasian FNR	IT	2	**Trust**
	UPDATE chain	UPDATE	2	pooled
	AA FNR §4.3	IT	2	**Trust**
	AA FPR gender	IUT	2	**UTrust**
	ProPublica headline	UPD.+IUT	3	**UTrust**
	Bayesian	I-P+E-P	2	posterior
	Racial FPR	IEX	2	Excess
	AA FNR gender	INEX	2	NoExcess
	+ 4 derivations with IUT2, IT2		2	varied
HMDA	Reference-rate	IT	2	**Trust**
	Temporal pool	UPD.+IUT2	3	**UTrust**
	Intersect. (M)	IUT2	2	**UTrust**
	Intersect. (F)	IUT2	2	**UTrust**
	Y-o-Y Black	IUT2	2	**UTrust**
	Y-o-Y White	IUT2	2	**UTrust**

Table 4
Lean-validated derivation trees (all pass `checkDerivation`).

6. Conclusion

We have presented TPTND-Lean, a Lean 4 implementation of a core TPTND fragment that type-checks derivation trees over real-world fairness data. The implementation adds two-sample trust rules (IT2/IUT2), specifies a four-stage workflow that takes raw group-level statistics through claim selection and derivation construction to a verified certificate, and exhibits the workflow on two public datasets spanning criminal justice and mortgage lending.

The main limitation is that the `Prob` type is clamped to $[0, 1]$, which loses information when the two-sample CI for a negative difference falls entirely below zero. In practice this is handled by ordering the premises so that the larger rate appears first. Future work includes signed-difference probability handling, connecting to the causal-label extensions of [3], and a front-end that constructs derivation trees from tabular data so domain experts can produce certificates without writing Lean.

Declaration on Generative AI

The first draft of this paper was written by the authors. During the subsequent preparation of this work, the authors used Claude (Anthropic) in order to: improve grammar and editorial polish; perform adversarial-style review of the paper and cross-checking of benchmark numbers against authoritative sources; and provide assistance with the Lean source code of the accompanying implementation. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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